Year 1 AS Pure Maths

Chapter 11

Vectors

Practice Booklet

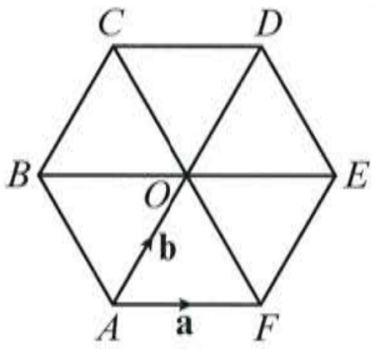
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11.1 Vectors

Practice Book, printed page 86–87.

1. $ABCDEF$ is a hexagon. $\overrightarrow{AF} = \mathbf{a}$ and $\overrightarrow{AO} = \mathbf{b}$.



Find in terms of **a** and **b**:

a. $\overrightarrow{AD}$ **b.** $\overrightarrow{EB}$ **c.** $\overrightarrow{CB}$

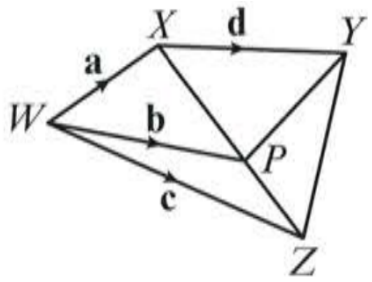
d. $\overrightarrow{AE}$ **e.** $\overrightarrow{OF}$ **f.** $\overrightarrow{FC}$

g. $\overrightarrow{CE}$ **h.** $\overrightarrow{BF}$

Hint

The vector $\overrightarrow{FA} = -\mathbf{a}$ and the vector $\overrightarrow{OA} = -\mathbf{b}$.

2. In the diagram, $\overrightarrow{WX} = \mathbf{a}$, $\overrightarrow{WP} = \mathbf{b}$, $\overrightarrow{WZ} = \mathbf{c}$ and $\overrightarrow{XY} = \mathbf{d}$.



Find in terms of **a**, **b**, **c** and **d**:

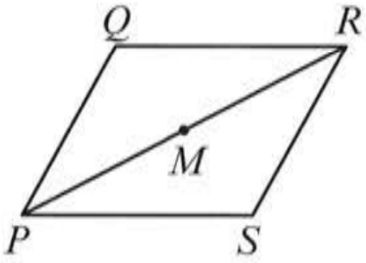
a. $\overrightarrow{XP}$ **b.** $\overrightarrow{PY}$ **c.** $\overrightarrow{PZ}$

d. $\overrightarrow{ZY}$

Hint

You can use answers to previous parts if it helps. For example, $\overrightarrow{PY} = \overrightarrow{PX} + \overrightarrow{XY}$.

3. In the parallelogram $PQRS$, $PQ = \mathbf{a}$ and $QR = \mathbf{b}$. The midpoint of PR is M.



Find in terms of **a** and **b**:

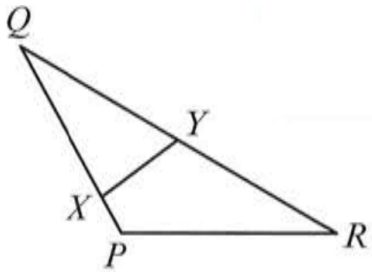
a. $\overrightarrow{PR}$ **b.** $\overrightarrow{PM}$ **c.** $\overrightarrow{SM}$

Hint

The midpoint M is halfway between P and R, so $\overrightarrow{PM} = \frac{1}{2}\overrightarrow{PR}$. ← Section 6.1

4. E The vector $8\mathbf{a} - 10\mathbf{b}$ is parallel to the vector $p\mathbf{a} + q\mathbf{b}$.
State, in simplest terms, the value of $\frac{p}{q}$. (1 mark)

5. E/P In the triangle PQR, $\overrightarrow{PQ} = \mathbf{a}$ and $\overrightarrow{PR} = \mathbf{b}$.
X divides PQ in the ratio $1 : 3$ and Y divides QR in the ratio $2 : 3$.

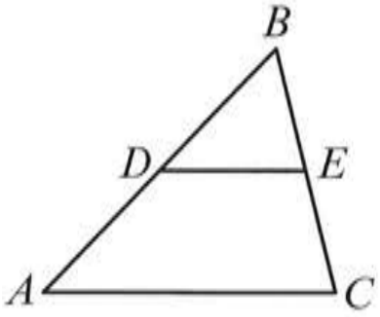


Find $\overrightarrow{YX}$ in terms of **a** and **b**. (4 marks)

Hint

If X divides PQ in the ratio $1 : 3$ then it is $\frac{1}{4}$ of the way along PQ from P.

6. E/P In the triangle ABC, $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{BC} = \mathbf{b}$.
D and E are the midpoints of AB and BC respectively.



Use vectors to prove that $\overrightarrow{AC}$ is parallel to $\overrightarrow{DE}$. (3 marks)

7. E/P $EFGH$ is a trapezium with FG parallel to EH and $\overrightarrow{FG} = k\overrightarrow{EH}$.
M is the midpoint of EF and N is the midpoint of GH.
$\overrightarrow{EH} = \mathbf{a}$, $\overrightarrow{EF} = \mathbf{b}$ and $\overrightarrow{HG} = \mathbf{c}$.

a. Prove that $\overrightarrow{MN} = \left(\frac{1+k}{2}\right)\mathbf{a}$. (3 marks)

b. Hence, or otherwise, explain why MN is parallel to EH and FG. (1 mark)

11.2 Representing vectors

Practice Book, printed page 87.

1. $\mathbf{a} = -7\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 2\mathbf{i} - 5\mathbf{j}$.
Find, in terms of $\mathbf{i}$ and $\mathbf{j}$:

a. $3\mathbf{a}$ **b.** $\mathbf{a} - 2\mathbf{b}$ **c.** $3\mathbf{b} - \mathbf{a}$

Hint

Add the $\mathbf{i}$ components and add the $\mathbf{j}$ components.

2. Given that $\mathbf{a} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -3 \\ -5 \end{pmatrix}$, find, as a column vector:

a. $\frac{1}{2}\mathbf{a}$ **b.** $\mathbf{a} + \mathbf{b} + \mathbf{c}$ **c.** $\mathbf{c} + 2\mathbf{b} - \frac{3}{2}\mathbf{a}$

Hint

$\begin{pmatrix} -2 \\ 4 \end{pmatrix}$ is another way of writing $-2\mathbf{i} + 4\mathbf{j}$.

3. Find the resultant of the vectors $\mathbf{a} = -6\mathbf{i} + 7\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - 5\mathbf{j}$.

Hint

The resultant is the vector sum of two or more vectors.

4. E Given that $\overrightarrow{AB} = -8\mathbf{i} + 7\mathbf{j}$ and $\overrightarrow{AC} = 11\mathbf{i} - \mathbf{j}$, find $\overrightarrow{BC}$. (2 marks)

5. E $\mathbf{a} = \begin{pmatrix} 2 \\ p \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -3 \\ -4 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} q \\ 11 \end{pmatrix}$
Given that $2\mathbf{b} - \mathbf{c} = \mathbf{a}$, find the value of p and the value of q. (3 marks)

6. E/P $\mathbf{a} = -2\mathbf{i} + 7\mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j}$, $\mathbf{c} = -\mathbf{i} - 6\mathbf{j}$ and $\mathbf{d} = 2\mathbf{i} - 2\mathbf{j}$.
Show that the vector $2\mathbf{a} + 5\mathbf{b}$ is not parallel to the vector $2\mathbf{d} - \mathbf{c}$. (3 marks)

7. E/P $\mathbf{a} = 2p\mathbf{i} + 3q\mathbf{j}$ and $\mathbf{b} = -4q\mathbf{i} + p\mathbf{j}$.
Given that $\mathbf{b} - \mathbf{a} = -10\mathbf{i} + 10\mathbf{j}$, find the value of p and the value of q. (3 marks)

11.3 Magnitude and direction

Practice Book, printed page 87–88.

1. Find the exact magnitude of the following vectors.

a. $6\mathbf{i} - 8\mathbf{j}$ **b.** $5\mathbf{i} + 4\mathbf{j}$ **c.** $-3\mathbf{i} + 6\mathbf{j}$

d. $-8\mathbf{i} - 8\mathbf{j}$

Hint

The magnitude of the vector $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ is $|\mathbf{a}| = \sqrt{x^2 + y^2}$. If you are asked to find the exact magnitude, leave your answer in simplified surd form. ← Section 1.5

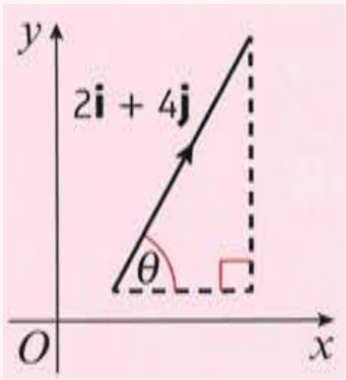
2. Find the angle that each of these vectors makes with the positive x-axis.

a. $2\mathbf{i} + 4\mathbf{j}$ **b.** $-7\mathbf{i} - \mathbf{j}$ **c.** $6\mathbf{j}$

d. $5\mathbf{i} - 5\mathbf{j}$

Hint

Draw a diagram and calculate the angle θ.

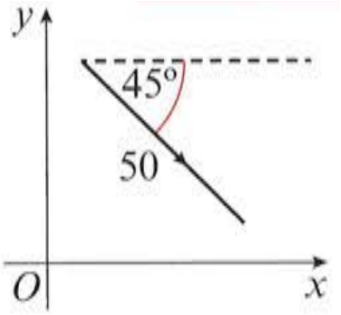


3. Find the unit vector in the direction $-5\mathbf{i} + 12\mathbf{j}$.

Hint

The unit vector in the direction of $\mathbf{a}$ is $\frac{\mathbf{a}}{|\mathbf{a}|}$.

4. Write the vector shown in the diagram in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are exact constants to be found.



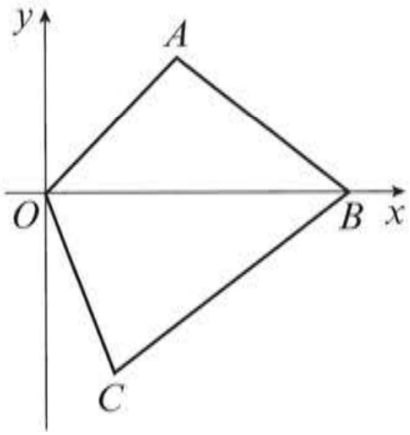
Hint

Use trigonometry to find the lengths of the x- and y-components of the vector.

5. E/P The vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$, where p and q are positive constants, is such that $|\mathbf{a}| = 11$. Given that $\mathbf{a}$ makes an angle of 20° with the positive y-axis, find the values of p and q. (2 marks)

6. E/P Given that $|2p\mathbf{i} + 3p\mathbf{j}| = 7\sqrt{13}$, find the possible values of p. (3 marks)

7. E Given that $\overrightarrow{OA} = 5\mathbf{i} + 4\mathbf{j}$, $\overrightarrow{AB} = 7\mathbf{i} - 4\mathbf{j}$ and $\overrightarrow{CB} = 10\mathbf{i} + 7\mathbf{j}$, find:



a. $\angle AOB$ (2 marks) **b.** $\angle BOC$ (3 marks) **c.** $\angle AOC$ (1 mark)

8. E/P Vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$ has magnitude 6.5 and makes an angle θ with the positive x-axis where $\tan\theta = -\frac{5}{12}$.
Find the possible values of p and q. (4 marks)

11.4 Position vectors

Practice Book, printed page 88–89.

1. The points P and Q have coordinates $(-2, 3)$ and $(7, 0)$ respectively.
Find, in terms of **i** and **j**:
a. the position vector of P **b.** the position vector of Q **c.** the vector $\overrightarrow{PQ}$.

Hint

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ where $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are the position vectors of P and Q respectively.

2. The coordinates of the triangle ABC, relative to the origin O, are $A(6, -2)$, $B(-1, -1)$ and $C(1, 3)$.
Find, in terms of **i** and **j**:
a. $\overrightarrow{OA}$ **b.** $\overrightarrow{BC}$ **c.** $\overrightarrow{CA}$

Hint

$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB}$ where $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are the position vectors of B and C respectively.

3. $\overrightarrow{OD} = -6\mathbf{i} + 8\mathbf{j}$ and $\overrightarrow{ED} = -9\mathbf{i} + 3\mathbf{j}$. Find, in surd form:
a. $|\overrightarrow{OD}|$ **b.** $|\overrightarrow{DE}|$ **c.** $|\overrightarrow{OE}|$

Hint

Use the given vectors to calculate $\overrightarrow{OE}$.

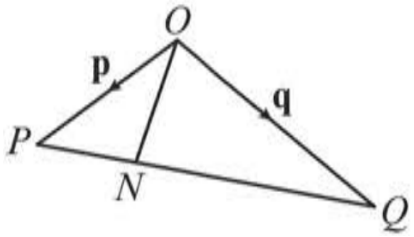
4. E $\overrightarrow{OA} = \begin{pmatrix} -7 \\ -6 \end{pmatrix}$ and $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 11 \end{pmatrix}$
Find:
a. the position vector of B (1 mark)
b. the exact value of $|\overrightarrow{OB}|$, giving your answer in surd form. (2 marks)

5. E/P The point P lies on the line with equation $y = 2x - 3$. Given that $|OP| = \sqrt{53}$, find possible expressions for $\overrightarrow{OP}$ in the form $p\mathbf{i} + q\mathbf{j}$. (4 marks)

11.5 Solving geometric problems

Practice Book, printed page 89–90.

1. Relative to a fixed origin O, the points P and Q have position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively. The point N divides PQ in the ratio $1 : 3$.



a. Write down $\overrightarrow{PQ}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

b. Copy and complete this expression by filling the blank space: $\overrightarrow{PN} = ___\overrightarrow{PQ}$

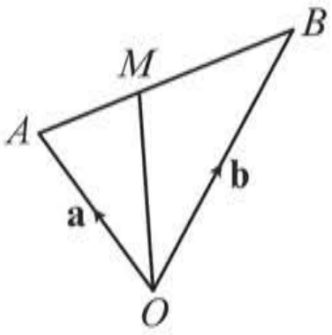
c. Copy and complete this expression by filling the blank space: $\overrightarrow{ON} = \overrightarrow{OP} + _____$

d. Find the position vector of N.

Hint

Use the ratio to work out what fraction of $\overrightarrow{PQ}$ is represented by $\overrightarrow{PN}$.

2. Relative to a fixed origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point M divides AB in the ratio $2 : 3$.

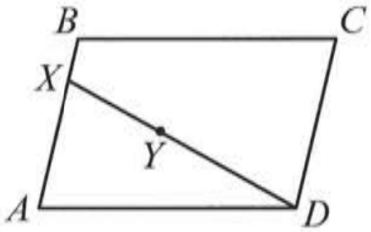


Find the position vector of M in terms of $\mathbf{a}$ and $\mathbf{b}$.

Hint

If M divides AB in the ratio $2 : 3$ then it is $\frac{2}{5}$ of the way along AB from A.

3. $ABCD$ is a parallelogram. $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{b}$. X divides AB in the ratio $3 : 1$ and Y divides XD in the ratio $2 : 3$.



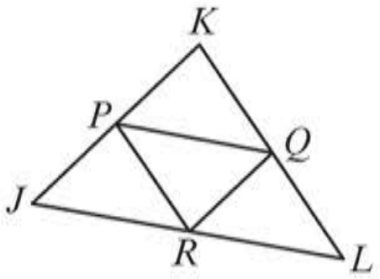
Find $\overrightarrow{XY}$.

Hint

First find $\overrightarrow{XD}$.

4. E In triangle PQR, $\overrightarrow{PQ} = 8\mathbf{i} + \mathbf{j}$ and $\overrightarrow{PR} = 2\mathbf{i} - 6\mathbf{j}$.
Find the size of $\angle QPR$ in degrees, correct to one decimal place. (5 marks)

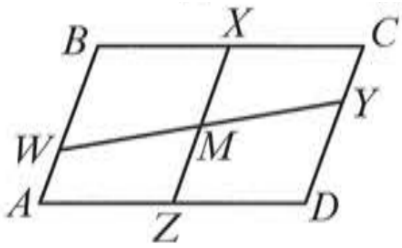
5. E/P Points P, Q and R are the midpoints of JK, KL and JL respectively.
Let $\overrightarrow{JK} = \mathbf{a}$ and $\overrightarrow{JL} = \mathbf{b}$.



a. Prove that $\overrightarrow{PR} = \frac{1}{2}\overrightarrow{KL}$. (3 marks)

b. Explain how you know that $\overrightarrow{PR}$ and $\overrightarrow{KL}$ must be parallel. (1 mark)

6. E/P $ABCD$ is a parallelogram. Let $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{b}$. Points X and Z are the midpoints of BC and AD respectively. Points W and Y divide AB and CD in the ratio $1 : 2$ respectively.



a. Find:

i. $\overrightarrow{WY}$

ii. $\overrightarrow{XZ}$

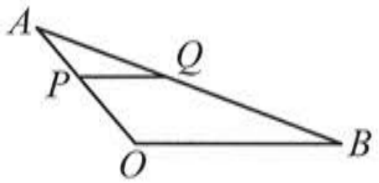
(2 marks)

b. Given that $\overrightarrow{WM} = \lambda\overrightarrow{WY}$, where λ is a constant, show that $\overrightarrow{AM} = \frac{1}{3}(1+\lambda)\mathbf{a} + \lambda\mathbf{b}$. (2 marks)

c. Given also that $\overrightarrow{XM} = \mu\overrightarrow{XZ}$, where μ is a constant, find an expression for $\overrightarrow{AM}$ in terms of a, b, and μ. (2 marks)

d. Hence, or otherwise, prove that M is the midpoint of WY and XZ. (4 marks)

7. E/P OAB is a triangle where P divides OA in the ratio $1 : k$ and Q divides BA in the ratio $1 : k$.



a. Prove that $\overrightarrow{PQ}$ is parallel to $\overrightarrow{OB}$. (4 marks)

b. Given that $|\overrightarrow{PQ}| = \frac{1}{3}|\overrightarrow{OB}|$, find the value of k. (2 marks)

11.6 Modelling with vectors

Practice Book, printed page 90–91.

1. Find the speed of a particle moving with these velocities:
a. $(5\mathbf{i} - 12\mathbf{j})$ m s^{-1} **b.** $(-6\mathbf{i} - 8\mathbf{j})$ km s^{-1} **c.** $(6\mathbf{i} + \mathbf{j})$ mm s^{-1}
d. $14\mathbf{j}$ cm s^{-1}

Hint

The speed of a particle travelling with velocity $(x\mathbf{i} + y\mathbf{j})$ m s^{-1} is $\sqrt{x^2 + y^2}$ m s^{-1}.

2. A particle moves with velocity $(-4\mathbf{i} + 3\mathbf{j})$ m s^{-1}. Find:
a. the speed of the particle
b. the distance travelled by the particle in 4 seconds
c. the time taken for the particle to travel 27.5 m.

Hint

Distance travelled = speed × time

3. At time $t = 0$, a particle P moves with velocity $(-7\mathbf{i} - 24\mathbf{j})$ km s^{-1}. The particle accelerates constantly such that, at time $t = 6$, it is moving with velocity $(2\mathbf{i} - 51\mathbf{j})$ km s^{-1}.
The acceleration vector of the particle is given by the formula $\mathbf{a} = \frac{\mathbf{v}-\mathbf{u}}{t}$
Find the acceleration of P in terms of $\mathbf{i}$ and $\mathbf{j}$.

4. Initially a particle is moving with velocity $(\mathbf{i} - 4\mathbf{j})$ m s^{-1}. The particle then accelerates with acceleration $(2\mathbf{i} - 3\mathbf{j})$ m s^{-2} for 4 seconds.
The velocity, $\mathbf{v}$, of the particle after t seconds is given by $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, where $\mathbf{u}$ is the initial velocity and $\mathbf{a}$ is the acceleration. Find:
a. the velocity of the particle after 4 seconds
b. the speed of the particle after 4 seconds.

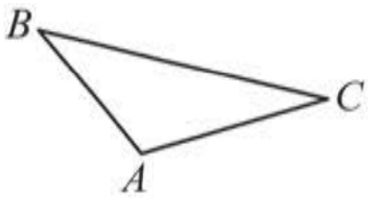
Hint

Substitute the given values of $\mathbf{u}$ and $\mathbf{a}$ into the formula $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ and then simplify your answer.

5. E Town A is 3 km west and 6 km north a fixed point O.
Town B is 8 km east and 2 km north of O.
a. Write down the position vector of A relative to O. (1 mark)
Find:
b. $|\overrightarrow{AB}|$ (3 marks)
c. the bearing of B from A. (3 marks)

6. E The forces $\mathbf{F}_1$ and $\mathbf{F}_2$ are given by the vectors $\mathbf{F}_1 = (p\mathbf{i} + q\mathbf{j})$ N and $\mathbf{F}_2 = (3\mathbf{i} - 7\mathbf{j})$ N. The resultant force $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$ acts in a direction parallel to the vector $(2\mathbf{i} - 3\mathbf{j})$.
a. Find the angle between $\mathbf{R}$ and the positive x-axis. (2 marks)
b. Show that $3p + 2q = 5$. (3 marks)
c. Given that $p = 3$, find the magnitude of $\mathbf{R}$. (2 marks)

7. E/P A forest fire starts at point A. The fire spreads and covers the area shown by the triangle in the diagram.



The vector $\overrightarrow{AB} = (-3\mathbf{i} + 5\mathbf{j})$ km and the vector $\overrightarrow{AC} = (6\mathbf{i} + \mathbf{j})$ km.
Find the area of forest which is on fire. (7 marks)

8. E A particle P moves under the action of two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (a\mathbf{i} + b\mathbf{j})$ N and $\mathbf{F}_2 = (-3\mathbf{i} + 7\mathbf{j})$ N. The resultant force acting on the particle is $\mathbf{R} = (8\mathbf{i} + 32\mathbf{j})$ N.

a. Find, to 1 decimal place, the angle between the resultant force and $\mathbf{i}$. (2 marks)

b. Find the value of a and the value of b. (3 marks)

A third force $\mathbf{F}_3 = (c\mathbf{i} + 10\mathbf{j})$ N is also applied to P, such that the resultant of all three forces acts parallel to $\mathbf{j}$.

c. Find the value of c. (3 marks)

Problem set A

Bronze

In triangle ABC, $\overrightarrow{AB} = 5\mathbf{i} + 12\mathbf{j}$ and $\overrightarrow{AC} = 12\mathbf{i} + 5\mathbf{j}$. Find:

a. $\overrightarrow{BC}$ (1 mark)

b. the lengths AB, AC and BC (3 marks)

c. $\angle BAC$ (2 marks)

Silver

The position vectors of the vertices in triangle ABC are $\overrightarrow{OA} = -\mathbf{i} + 4\mathbf{j}$, $\overrightarrow{OB} = 8\mathbf{j}$ and $\overrightarrow{OC} = 7\mathbf{i} + 9\mathbf{j}$. Find:

a. $\overrightarrow{BC}$ (1 mark)

b. $\angle BAC$ (4 marks)

c. the area of triangle ABC. (2 marks)

Gold

In triangle ABC, $\overrightarrow{AB} = 8\mathbf{i} - 2\mathbf{j}$ and $\overrightarrow{AC} = 2\mathbf{i} - 7\mathbf{j}$. Find the area of triangle ABC. (7 marks)

Problem set B

Bronze

In triangle OAB, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. R divides AB in the ratio $3 : 2$. $\overrightarrow{OP} = 5\overrightarrow{OR}$.

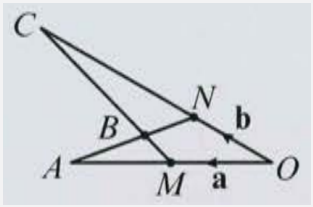
a. Show that $\overrightarrow{OP} = 2\mathbf{a} + 3\mathbf{b}$. (3 marks)

Point Q is such that $\overrightarrow{OQ} = -\mathbf{a}$.

b. Prove that the points Q, B and P are collinear. (3 marks)

Silver

In the diagram, M is the midpoint of OA and N lies on OC. The point B lies on MC such that $MB : BC = 1 : 4$. $\overrightarrow{OM} = \mathbf{a}$ and $\overrightarrow{ON} = \mathbf{b}$. Given that $\overrightarrow{AB} = k\overrightarrow{AN}$,

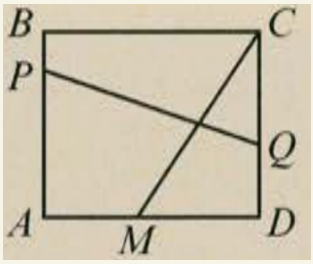


a. find the value of k (4 marks)

b. write $\overrightarrow{OC}$ as a scalar multiple of $\mathbf{b}$. (1 mark)

Gold

$ABCD$ is a square. P divides AB in the ratio $4 : 1$ and Q divides CD in the ratio $3 : 2$. M is the midpoint of AD. Prove that PQ bisects MC. (6 marks)



Selected exam question bank questions

Question 5

Given that $\mathbf{a} = 6\mathbf{i} - 3\mathbf{j}$, find:

a. the unit vector in the same direction of $\mathbf{a}$ (3 marks)

b. the angle that $\mathbf{a}$ makes with the positive x-axis. (2 marks)

Question 14

The resultant of the forces $\mathbf{F}_1 = 6\mathbf{i} + 3p\mathbf{j}$ and $\mathbf{F}_2 = 4p\mathbf{i} - 5\mathbf{j}$ acts parallel to $\mathbf{i}$. Find:

a. the value of p (2 marks)

b. the magnitude of the resultant force. (2 marks)

Question 16

At time $t = 0$, a ship, S, and a boat, B, sail from position O. The ship sails with velocity vector $(-\mathbf{i} + 3\mathbf{j})$ km h^{-1} and the boat sails with velocity vector $(6\mathbf{i} - 2\mathbf{j})$ km h^{-1}.

a. Work out the position vector of the ship and the boat after 2 hours. (2 marks)

b. Calculate the distance of the ship from the boat after 2 hours, leaving your answer as a simplified surd. (3 marks)

Question 19

The point B lies on the line with equation $3x + 2y = 12$. Given that $|OB| = 10$, find the vector $\overrightarrow{OB}$ in the form $p\mathbf{i} + q\mathbf{j}$, where $p > 0$. (4 marks)

Question 56

In the triangle DEF, $\overrightarrow{DE} = 3\mathbf{i} + 7\mathbf{j}$ and $\overrightarrow{DF} = 8\mathbf{i} - \mathbf{j}$.

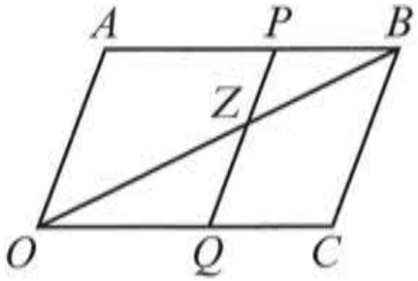
a. Prove that triangle DEF is scalene. (4 marks)

b. Find the size of $\angle DFE$, correct to 1 decimal place. (4 marks)

Question 80

$OABC$ is a parallelogram. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{b}$.

The point Z divides OB in the ratio $3 : 2$.



a. Express the vector $\overrightarrow{CZ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. (2 marks)

b. Explain whether the straight line which passes through C and Z will also pass through the point A. (2 marks)

Given that OA, QP and CB are parallel,

c. show that $AP : PB = 3 : 2$ and $PZ : ZQ = 2 : 3$. (3 marks)

Answers

11.1 Vectors

1. **a.** $2\mathbf{b}$
   b. $-2\mathbf{a}$
   c. $-\mathbf{b}$
   d. $\mathbf{a}+\mathbf{b}$
   e. $\mathbf{a}-\mathbf{b}$
   f. $-2\mathbf{a}+2\mathbf{b}$
   g. $2\mathbf{a}-\mathbf{b}$
   h. $2\mathbf{a}-\mathbf{b}$
2. **a.** $-\mathbf{a}+\mathbf{b}$
   b. $\mathbf{a}-\mathbf{b}+\mathbf{d}$
   c. $-\mathbf{b}+\mathbf{c}$
   d. $\mathbf{a}-\mathbf{c}+\mathbf{d}$
3. **a.** $\mathbf{a}+\mathbf{b}$
   b. $\frac{1}{2}\mathbf{a}+\frac{1}{2}\mathbf{b}$
   c. $\frac{1}{2}\mathbf{a}-\frac{1}{2}\mathbf{b}$
4. $-\frac{4}{5}$
5. $-\frac{7}{20}\mathbf{a}-\frac{2}{5}\mathbf{b}$
6. $\overrightarrow{AC}=\mathbf{a}+\mathbf{b}$, $\overrightarrow{DE}=\frac{1}{2}\mathbf{a}+\frac{1}{2}\mathbf{b}=\frac{1}{2}\overrightarrow{AC}$, so the two lines are parallel as one vector is a scalar multiple of the other.
7. **a.** $\overrightarrow{MN}=\frac{1}{2}\mathbf{b}+k\mathbf{a}-\frac{1}{2}\mathbf{c}$ and $\overrightarrow{MN}=-\frac{1}{2}\mathbf{b}+\mathbf{a}+\frac{1}{2}\mathbf{c}$
   Adding gives $2\overrightarrow{MN}=k\mathbf{a}+\mathbf{a}$, so $\overrightarrow{MN}=\left(\frac{1+k}{2}\right)\mathbf{a}$
   b. $\overrightarrow{EH}$, $\overrightarrow{FG}$ and $\overrightarrow{MN}$ are all scalar multiples of $\mathbf{a}$, so are parallel to each other.

11.2 Representing vectors

1. **a.** $-21\mathbf{i}+9\mathbf{j}$
   b. $-11\mathbf{i}+13\mathbf{j}$
   c. $13\mathbf{i}-18\mathbf{j}$
2. **a.** $\begin{pmatrix}-1\\2\end{pmatrix}$
   b. $\begin{pmatrix}1\\-2\end{pmatrix}$
   c. $\begin{pmatrix}12\\-13\end{pmatrix}$
3. $-3\mathbf{i}+2\mathbf{j}$
4. $19\mathbf{i}-8\mathbf{j}$
5. $p=-19$, $q=-8$
6. $2\mathbf{a}+5\mathbf{b}=11\mathbf{i}+4\mathbf{j}$, $2\mathbf{d}-\mathbf{c}=5\mathbf{i}+2\mathbf{j}$, so the vectors are not parallel.
7. $p=7$, $q=-1$

11.3 Magnitude and direction

1.
- **a.** 10
- **b.** $\sqrt{41}$
- **c.** $3\sqrt{5}$
- **d.** $8\sqrt{2}$

2.
- **a.** 63.4° above (1 d.p.)
- **b.** 171.9° below (1 d.p.)
- **c.** 90° above
- **d.** 45° below

3. $-\frac{5}{13}\mathbf{i} + \frac{12}{13}\mathbf{j}$

4. $25\sqrt{2}\mathbf{i} - 25\sqrt{2}\mathbf{j}$

5. $p = 3.76$, $q = 10.3$ (3 s.f.)

6. $p = \pm 7$

7.
- **a.** 38.7° (1 d.p.)
- **b.** 74.1° (1 d.p.)
- **c.** 112.7° (1 d.p.)

8. $p = \pm 6$ and $q = \mp 2.5$

11.4 Position vectors

1.
- **a.** $-2\mathbf{i} + 3\mathbf{j}$
- **b.** $7\mathbf{i}$
- **c.** $9\mathbf{i} - 3\mathbf{j}$

2.
- **a.** $6\mathbf{i} - 2\mathbf{j}$
- **b.** $2\mathbf{i} + 4\mathbf{j}$
- **c.** $5\mathbf{i} - 5\mathbf{j}$

3.
- **a.** 10
- **b.** $3\sqrt{10}$
- **c.** $\sqrt{34}$

4.
- **a.** $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$
- **b.** $\sqrt{41}$

5. $\frac{22}{5}\mathbf{i} + \frac{29}{5}\mathbf{j}$ and $-2\mathbf{i} - 7\mathbf{j}$

11.5 Solving geometric problems

1.
- **a.** $\mathbf{q} - \mathbf{p}$
- **b.** $\overrightarrow{PN} = \frac{1}{4}\overrightarrow{PQ}$
- **c.** $\overrightarrow{ON} = \overrightarrow{OP} + \overrightarrow{PN}$
- **d.** $\frac{3}{4}\mathbf{p} + \frac{1}{4}\mathbf{q}$

2. $\frac{3}{5}\mathbf{a} + \frac{2}{5}\mathbf{b}$

3. $\overrightarrow{XY} = -\frac{3}{10}\mathbf{a} + \frac{2}{5}\mathbf{b}$

4. 78.7°

5.
- **a.** $\overrightarrow{KL} = \mathbf{b} - \mathbf{a}$, $\overrightarrow{PR} = \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}$, $\overrightarrow{PR} = \frac{1}{2}\overrightarrow{KL}$
- **b.** Any two vectors, where one is a scalar multiple of the other, are parallel.

6.
- **a.**
  - **i.** $\frac{1}{3}\mathbf{a} + \mathbf{b}$
  - **ii.** $-\mathbf{a}$
- **b.** $\overrightarrow{AM} = \overrightarrow{AW} + \overrightarrow{WM} = \overrightarrow{AW} + \lambda\overrightarrow{WY} = \frac{1}{3}\mathbf{a} + \lambda\left(\frac{1}{3}\mathbf{a} + \mathbf{b}\right)$
  $= \frac{1}{3}(1+\lambda)\mathbf{a} + \lambda\mathbf{b}$
- **c.** $\overrightarrow{AM} = \overrightarrow{AB} + \overrightarrow{BX} + \overrightarrow{XM} = \mathbf{a} + \frac{1}{2}\mathbf{b} - \mu\mathbf{a} = (1-\mu)\mathbf{a} + \frac{1}{2}\mathbf{b}$
- **d.** Equating coefficients of $\mathbf{a}$ and $\mathbf{b}$, $\mu + \frac{1}{3}\lambda = \frac{2}{3}$ and $\lambda = \frac{1}{2}$
  So $\mu = \frac{2}{3} - \frac{1}{6} = \frac{1}{2} \Rightarrow M$ is the midpoint of WY and XZ.

7.
- **a.** $\overrightarrow{PQ} = \frac{k}{1+k}\overrightarrow{OA} - \frac{k}{1+k}\overrightarrow{BA} = \frac{k}{1+k}\overrightarrow{OB}$. So $\overrightarrow{PQ}$ is a scalar multiple of $\overrightarrow{OB}$ and these vectors are parallel.
- **b.** $k = \frac{1}{2}$

11.6 Modelling with vectors

1.
- **a.** 13 m s^{-1}
- **b.** 10 km s^{-1}
- **c.** $\sqrt{37}$ mm s^{-1}
- **d.** 14 cm s^{-1}

2.
- **a.** 5 m s^{-1}
- **b.** 20 m
- **c.** 5.5 seconds

3. $(\frac{3}{2}\mathbf{i} - \frac{9}{2}\mathbf{j})$ km s^{-2}

4.
- **a.** $(9\mathbf{i} - 16\mathbf{j})$ m s^{-1}
- **b.** $\sqrt{337}$ m s^{-1}

5.
- **a.** $(-3\mathbf{i} + 6\mathbf{j})$ km
- **b.** $\sqrt{137}$ km
- **c.** 110.0° (1 d.p.)

6.
- **a.** 56.3° (1 d.p.) below the positive x-axis
- **b.** Solve $3 + p = 2k$ and $q - 7 = -3k$ by eliminating k.
- **c.** $3\sqrt{13}$ N

7. 16.5 km^2

8.

a. $76.0°$

b. $a = 11$, $b = 25$

c. $c = -8$

Problem-solving Set A

Bronze

a. $\mathbf{7i - 7j}$

b. $AB = 13$, $AC = 13$, $BC = 7\sqrt{2}$

c. $44.8°$

Silver

a. $\mathbf{7i + j}$

b. $44.0°$

c. 13.5

Gold 26

Problem-solving Set B

Bronze

a. $\overrightarrow{OR} = \overrightarrow{OA} + \overrightarrow{AR} = \mathbf{a} + \frac{3}{5}(-\mathbf{a} + \mathbf{b}) = \frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}$
$\overrightarrow{OP} = 5\overrightarrow{OR} = 5(\frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}) = 2\mathbf{a} + 3\mathbf{b}$

b. $\overrightarrow{QB} = \mathbf{a} + \mathbf{b}$, $\overrightarrow{BP} = 2\mathbf{a} + 2\mathbf{b}$, $\overrightarrow{QB} = \frac{1}{2}\overrightarrow{BP}$ so lines are parallel. As B is a shared point, Q, B and P are collinear.

Silver

a. $\frac{3}{5}$

b. $\overrightarrow{OC} = 3\mathbf{b}$

Gold Let $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{b}$. Let point of intersection of PQ and MC be R, and let $\overrightarrow{MR} = k\overrightarrow{MC}$ and $\overrightarrow{PR} = j\overrightarrow{PQ}$. So going via M,

$\overrightarrow{AR} = \frac{1}{2}\mathbf{b} + k(\frac{1}{2}\mathbf{b} + \mathbf{a}) = k\mathbf{a} + (\frac{1}{2}k + \frac{1}{2})\mathbf{b}$

And going via P,

$\overrightarrow{AR} = \frac{4}{5}\mathbf{a} + j(\mathbf{b} - \frac{2}{5}\mathbf{a}) = (\frac{4}{5} - \frac{2}{5}j)\mathbf{a} + j\mathbf{b}$

Equating coefficients of $\mathbf{a}$ and $\mathbf{b}$, $k = \frac{4}{5} - \frac{2}{5}j$ and $j = \frac{1}{2}k + \frac{1}{2}$ which gives $k = \frac{1}{2}$.

Selected exam question bank answers

Question 5

a. $\frac{2\sqrt{5}}{5}\mathbf{i} - \frac{\sqrt{5}}{5}\mathbf{j}$

b. $26.6°$ (1 d.p.) below the x-axis

Question 14

a. $p = \frac{5}{3}$

b. $\frac{38}{3}$

Question 16

a. Ship: $-2\mathbf{i} + 6\mathbf{j}$, boat: $12\mathbf{i} - 4\mathbf{j}$

b. $2\sqrt{74}$

Question 19 $8\mathbf{i} - 6\mathbf{j}$

Question 56

a. $|\overrightarrow{DE}| = \sqrt{58}$, $|\overrightarrow{EF}| = \sqrt{89}$ and $|\overrightarrow{DF}| = \sqrt{65}$, so triangle DEF is scalene.

b. $50.9°$

Question 80

a. $\frac{3}{5}\mathbf{a} - \frac{2}{5}\mathbf{b}$

b. $\overrightarrow{CA} = \mathbf{a} - \mathbf{b}$. As $\overrightarrow{CA}$ and $\overrightarrow{CZ}$ are not parallel, $\overrightarrow{CZ}$ will not pass through the point A.

c. Create a vector equation using $\overrightarrow{OZ} = \overrightarrow{OA} + \overrightarrow{AP} + \overrightarrow{PZ}$ and equate **a** and **b** parts.